

OpenCV Lab Assignment 1

Using Python and Google Colab

Prepared by: Nusha Yamina

ID: 231003

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Task 1

1. Reading any image.
2. Display images in RGB and Grayscale
3. Extract image dimensions
4. Access pixel values at specific locations
5. Crop a Region of Interest (ROI)
6. Draw rectangle and add text on the image

Load and Display Image

```
import cv2
import matplotlib.pyplot as plt
```

```
img = cv2.imread("your_image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("RGB Image")
plt.axis('off')
```

```
plt.subplot(1,2,2)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis('off')
plt.show()
```

RGB Image



Grayscale Image



Dimension and Pixel Values of Images

1. DIMENSIONS

```
print("Image Dimensions (H,W,C):", img.shape)
```

2. PIXEL VALUES

```
locations = [(10,10), (50,100), (120,201)]
```

```
for (x,y) in locations:
```

```
    print(f"Pixel at {x,y} in RGB:", img[x,y])
```

```
    print(f"Pixel at {x,y} in Gray:", gray[x,y])
```

```
Image Dimensions (H,W,C): (1008, 1008, 3)
Pixel at (10, 10) in RGB: [134 129 126]
Pixel at (10, 10) in Gray: 129
Pixel at (50, 100) in RGB: [131 129 128]
Pixel at (50, 100) in Gray: 129
Pixel at (120, 201) in RGB: [116 119 124]
Pixel at (120, 201) in Gray: 120
```

Finding Region of Interest (ROI) in Image

```
# 3. REGION OF INTEREST (ROI)

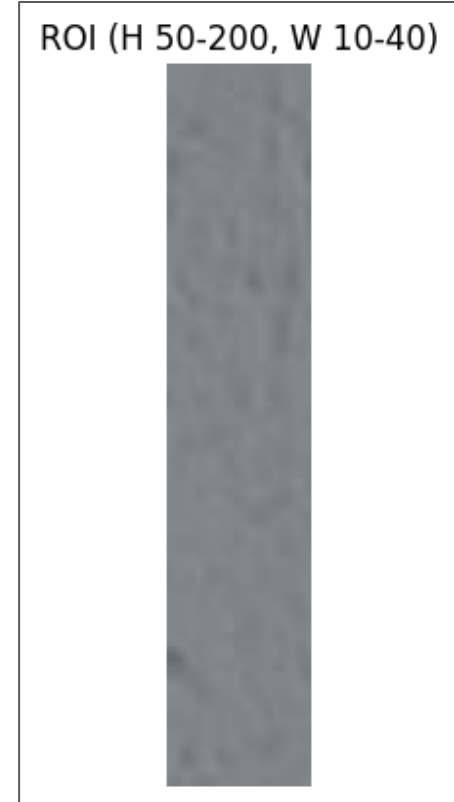
roi = img[50:200, 10:40]

plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))

plt.title("ROI (H 50-200, W 10-40)")

plt.axis('off')

plt.show()
```



Drawing Rectangle on Image

4. DRAWING RECTANGLE

```
img_rect = img.copy()
cv2.rectangle(img_rect, (250, 310),(570, 600), (0, 0, 255), 5)
plt.imshow(cv2.cvtColor(img_rect, cv2.COLOR_BGR2RGB))
plt.title("Rectangle Drawn")
plt.axis('off')
plt.show()
```



Writing Name & ID in Image

```
# 5. WRITING NAME & ID
```

```
img_text = img.copy()
```

```
cv2.putText(img_text, "Nusha Yamina~231003", (300,617),
```

```
cv2.FONT_HERSHEY_SIMPLEX, 1, (0,255,255), 3)
```

```
plt.imshow(cv2.cvtColor(img_text, cv2.COLOR_BGR2RGB))
```

```
plt.title("Text Written")
```

```
plt.axis('off')
```

```
plt.show()
```

Text Written



Task 2 : Creating Border

Image with Border



```
# FOR COLORED IMG
```

```
img = cv2.imread(CV)
```

```
h, w = img.shape[:2]
```

```
# CREATING BORDER
```

```
top = bottom = int(0.10 * h)
```

```
left = right = int(0.10 * w)
```

```
bordered = cv2.copyMakeBorder(
```

```
    img, top, bottom, left, right,
```

```
    cv2.BORDER_CONSTANT, value=[0, 255, 0] )
```

```
# Displaying IMG
```

```
plt.imshow(cv2.cvtColor(bordered, cv2.COLOR_BGR2RGB))
```

```
plt.title("Image with Border")
```

```
plt.axis('off')
```

```
plt.show()
```

```
# SAVE IMG
```

```
cv2.imwrite("bordered_output.jpg", bordered)
```

```
print("Saved as bordered_output.jpg")
```

```
files.download("bordered_output.jpg")
```

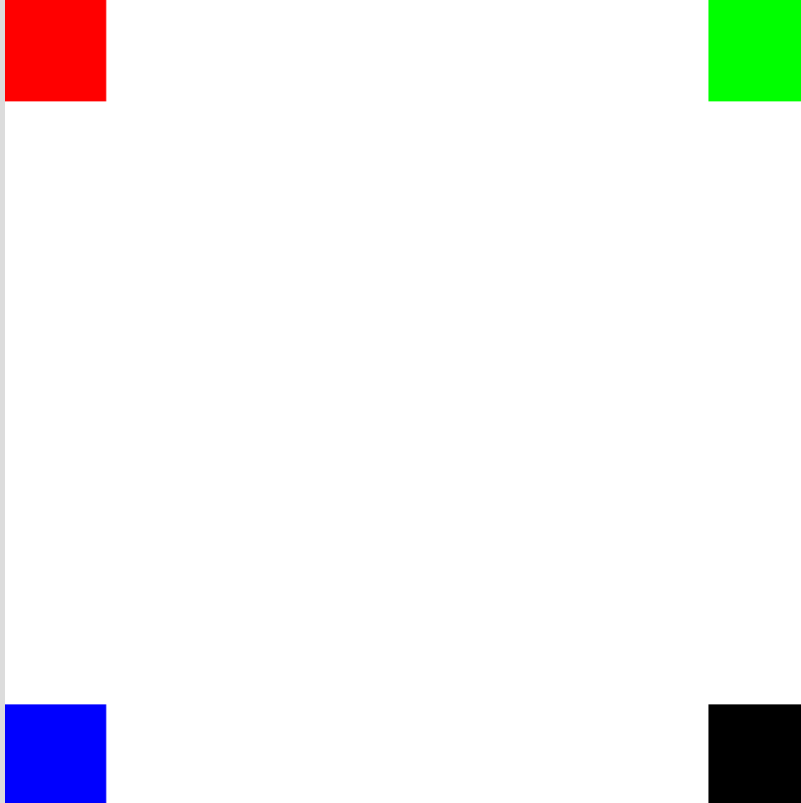
Task 2.2: For Greyscale Image

Grayscale Image with Border



```
# FOR GREYSCALE IMG
top = bottom = int(0.10 * h)
left = right = int(0.10 * w)
# CREATING BORDER
bordered_gray = cv2.copyMakeBorder(
    gray, top, bottom, left, right,
    cv2.BORDER_CONSTANT,
    value= 0 )
# Displaying IMG
plt.imshow(bordered_gray, cmap='gray')
plt.title("Grayscale Image with Border")
plt.axis('off')
plt.show()
# SAVE IMG
cv2.imwrite("bordered_gray.jpg", bordered_gray)
print("Saved as bordered_gray.jpg")
files.download("bordered_gray.jpg")
```

Task 3: Creating Image



```
import numpy as np
import cv2
from google.colab.patches import cv2_imshow

def create_corner_boxes(size=500):
    # White background
    img = np.ones((size, size, 3), dtype=np.uint8) * 255
    box = size // 8
    # Top-left: Red
    cv2.rectangle(img, (0, 0), (box, box), (0, 0, 255), -1)
    # Top-right: Green
    cv2.rectangle(img, (size - box, 0), (size, box), (0, 255, 0), -1)
    # Bottom-left: Blue
    cv2.rectangle(img, (0, size - box), (box, size), (255, 0, 0), -1)
    # Bottom-right: Black
    cv2.rectangle(img, (size - box, size - box), (size, size), (0, 0, 0), -1)
    cv2_imshow(img)
create_corner_boxes(500)
```